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EXPERIMENT 1

Aim: To study and prepare a summary of a research paper related to cognitive computing.

Research Paper Title: Artificial Intelligence In Medicine- Cognitive computing technological trends and future research directions in healthcare — A systematic literature review.

Summary: The paper provides a comprehensive survey of cognitive computing's applications and implications in healthcare, highlighting its transformative potential in diagnosing, treating, and managing diseases.

Cognitive computing, which mimics human reasoning and memory processes through advanced machine learning (ML) and artificial intelligence (AI) techniques, has become increasingly pivotal in various sectors, with healthcare being a primary focus. The introduction outlines how cognitive systems enhance decision-making capabilities for medical professionals by processing vast amounts of data—both structured and unstructured. It emphasizes the exponential growth of healthcare data, which complicates traditional data analysis methods, and proposes cognitive computing as a solution to derive meaningful insights.

The survey specifically examines recent developments and applications of cognitive computing technologies such as predictive analytics, automated reasoning, and personalized treatment plans. It discusses the significance of tools like the Clinical Trial Matching (CTM) system, which utilizes natural language processing (NLP) to assist physicians in identifying suitable clinical trials for cancer patients more efficiently.

Furthermore, the paper identifies current challenges in healthcare, including the complexity of medical data, time-consuming review processes, and inadequate human-computer interaction. It suggests future enhancements like cognitive handwriting recognition systems, automated knowledge ingestion, and intelligent cognitive medical assistants to facilitate better diagnosis and treatment.

Analytical synthesis of the literature reveals a growing trend in research output, particularly from the United States, with an increasing number of publications focusing on cognitive computing in healthcare over the years 2014 to 2021. The authors also outline the methodological approach based on PRISMA guidelines for systematic reviews, ensuring a rigorous assessment of the research literature included.

In conclusion, the paper asserts that cognitive computing, an evolving technology in healthcare augments the clinical thought process and enable the doctors to make the right diagnosis and preserve the patient's health in good condition. These systems provides timely care, optimal and cost-effective treatment. This article provides an extensive survey of the importance of cognitive computing in the health sector by highlighting the platforms, techniques, tools, algorithms, applications, and use cases. This survey also explores about the works in the literature on present issues and proposes the future research directions of applying cognitive systems in healthcare.